HUNGRY WOLF

Verified Points. Friendly Competition. Built to Extend.

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Repo
https://github.com/tmddnrdl333/CSC510_SE26F

Walkthrough Video
https://go.ncsu.edu/rh4swnv

Discussion Forum
https://discord.gg/jQj9gKQd

Mission Statement

Food delivery apps reward every customer the same, with no sense of progress and no one to share it with. Hungry Wolf turns ordering into a quest, but our Project 1a tests showed the numbers behind it could be forged. Project 2 fixes that and makes it social: a trust layer verifies every point, delivery, and donation; a leaderboard ranks friends by points, badges, and meals donated; and team challenges let friends earn donations together. Our M0 eval tests whether this lifts weekly repeat orders by at least 20%. **Result: +24.91% lift vs the 20% target (PASS, p = 0.015) on seeded simulated order logs, run in CI on every commit.**

Pick Us for Project 3

Ship on day one. A new team challenge is one definition object, with a test proving no engine changes are needed. A new metric is one small function.

Your four next milestones are already scoped. Seasons, caching, admin role, and M0 pilot testing!

Hard to break. 329 tests run, so you can pinpoint where something breaks.

Running in minutes. INSTALL.md, a Docker emulator, and a seed script with demo users ready to log in.

Stakeholders

Customers: compete with friends, chase badges, see donation impact that they can trust.

Friends and Teams: the social circle the leaderboard and the challenges are built for.

Delivery Partners: paid only for deliveries they actually completed.

Restaurants: correct 10% / 15% points make loyalty visible and easier to track.

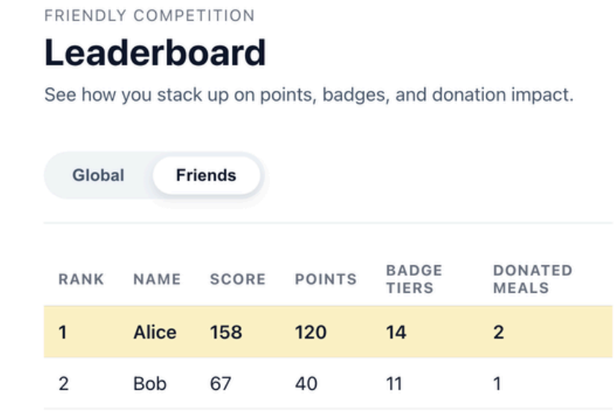
329 Tests, 325 Passing

Run tests with coverage

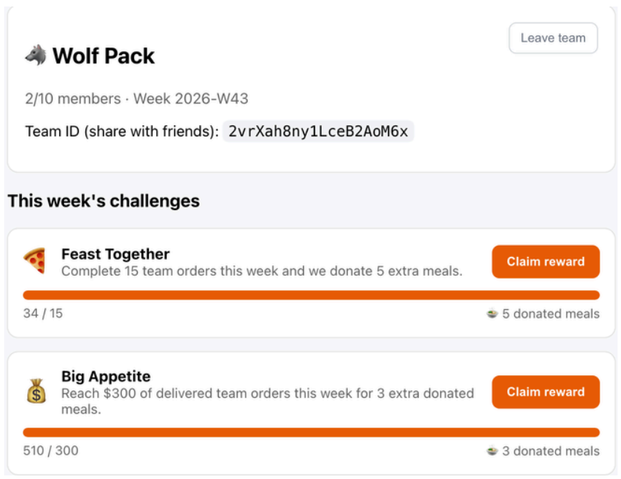
```
Test Suites: 4 failed, 27 passed, 31 total
Tests:       4 failed, 325 passed, 329 total
Snapshots:   0 total
Time:        10.362 s
Ran all test suites.
```

Run in CI on every commit. 78% coverage. The 4 failures are documented Project 1a bugs (#69 to #72).

What Works Today!



Friends leaderboard: score = points + 2 × badge tiers + 5 × meals



Team challenges: progress fills as friends order

```
------------------------------------------------------------------------------
METRIC SUMMARY:
  Baseline (No-Leaderboard) Repeat Rate :   34.10%  (133/390 active customer-weeks)
  Leaderboard Group Repeat Rate         :   42.60%  (164/385 active customer-weeks)
  Relative Lift                         :  +24.91%  (Target: >= +20.0%)
  Statistical Significance (z-test)     : z = 2.432, p = 0.0150
------------------------------------------------------------------------------
HYPOTHESIS VERDICT : [ PASS ]
PROJECT DECISION   : [ PERSEVERE ]
------------------------------------------------------------------------------
```

M0 eval: +24.91% lift, PASS (p = 0.015)

Milestones

Before: Project 1a

- **Order Lifecycle:** cart, pending, preparing, out for delivery, delivered.
- **Gamification:** 18 badges with 4 tiers each, 1 meal per 10 orders, points.
- **Voice Navigation:** 5 Gemini classified commands.
- **Test Baseline:** 120 tests, 20 of 20 use cases, which helped expose the trust gaps.

Now: Project 2, working

- **M0 Eval:** repeat-order-rate instrument on seeded simulated logs, run in CI. Result: +24.91%, PASS.
- **Trust Layer:** delivery credit only for the assigned partner, transaction-safe redemption, points at 10%/15% of the bill.
- **Social Leaderboard:** score = points + 2 x badge tiers + 5 x meals donated; Global and Friends views.
- **Team challenges:** weekly team goals; rewards paid exactly once via a Firestore transaction.

Future, Project 3

- **Seasons:** weekly/monthly leaderboard resets and seasonal badge tiers, via a reset job and a season field.
- **Scale:** cached global leaderboard snapshot.
- **Admin Role:** stop restaurants from self-granting Local Legend status, require roles.
- **Run M0 on Pilot Users:** export real logs from Firestore and point m0-eval at them.

Tech Stack: React, TS, Firebase, Express, JS

Inherit this to learn JWT auth, Firestore transactions, and rule-driven engines.